PRODUCT MANAGEMENT CASE STUDY

AI Retention Co-Pilot

Strategic AI Intervention to Combat Behavioral Decay and Protect **₹90L** in Annual Recurring Revenue.

SECTOR	ARR	KEY IMPACT	USER BASE
B2B SaaS	₹5 Crore	₹7.5L Saved/Mo	50,000+ Active Users

01. EXECUTIVE SUMMARY

The Mission

The goal was to transform a reactive "Firefighting" Customer Success culture into a proactive, data-driven retention engine. We moved from tracking *who cancelled* to predicting *who will cancel*.

Stakeholder Alignment: Secured buy-in from C-Suite by demonstrating the LTV:CAC impact of a 1.5% churn reduction. Focus areas included Behavioral Modeling and Revenue Ops integration.

> *"The breakthrough happened when we realized churn isn't a single event—it's a 21-day decay of user habits that we were simply ignoring."*

7.0%

INITIAL CHURN

5.5%

TARGET CHURN

₹90,00,000

ANNUAL REVENUE PROTECTED

Deployment: Q3 Pilot Phase | 40 Accounts Sampled

02. PROBLEM DISCOVERY

The Status Quo: A Flawed Workflow

Customer Success Managers were blind to behavioral shifts. They managed 50+ accounts each, relying on "Lagging Indicators" that only signaled trouble when it was too late:

- **NPS Surveys:** Sent quarterly, missed by the users already disengaging.
- **Renewal Conversations:** Often the first time value was discussed in 6 months.
- **Support Tickets:** Viewed as isolated issues, not symptoms of decay.

Cost of Inaction: Compounding churn led to a projected 15% drop in total fiscal year ARR if left unaddressed. Firefighting consumed 80% of CSM bandwidth.

THE BEHAVIORAL AUDIT

Research Methodology

To move beyond guesswork, I conducted a deep-dive retrospective audit of **120 accounts** churned over the last 6 months. I cross-referenced three data silos:

Product Logs

Feature clicks & login depth.

Zendesk

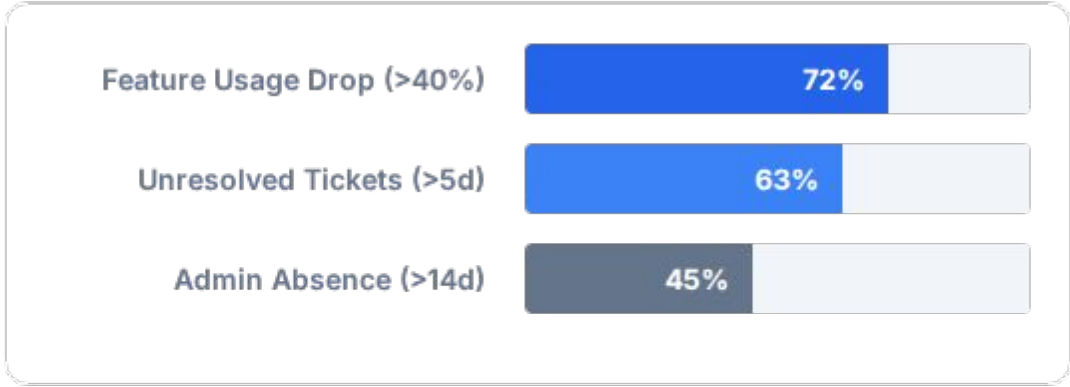
Ticket friction & sentiment.

HubSpot

CSM notes & engagement.

Sample Context: 4.2 Million individual event logs analyzed over a 180-day window. Identified 12 distinct behavioral archetypes across 4 market tiers.

QUANTITATIVE AUDIT RESULTS



The Signal: When a Primary Admin stops logging in for 2 weeks, the risk of churn increases by 4x even if end-users are active.

Confidence Interval: 95% | Data Sourced from Snowflake Data Warehouse

03. USER PERSONAS

Arjun

OVERWHELMED CSM

"I manage 50 accounts. I don't have time for 'audits'. I just need to know who is about to leave before they stop replying to my emails."

Pain Point: Decision fatigue & high-priority noise.

Desired Outcome: Zero-effort daily prioritization list.

Current Tech: Salesforce, Intercom, Gong

Priya

STRATEGIC CS DIRECTOR

"I need a predictable way to forecast churn so I can justify our CS headcount to the board based on revenue protection."

Pain Point: Zero visibility into portfolio risk.

Desired Outcome: Quantifiable NRR (Net Revenue Retention) growth.

Current Tech: Excel, Tableau, Salesforce

04. PRODUCT ARCHITECTURE

The Proprietary Health Score

We moved away from binary "Active/Inactive" statuses to a weighted **Predictive Health Score (0–100)**. This score reflects the velocity of behavioral change, not just a static snapshot.

$$\text{Health} = (0.5 \cdot U) + (0.3 \cdot S) + (0.2 \cdot E)$$

Signal Validation: Noise reduction algorithms applied to U (Usage) to filter out occasional usage spikes and seasonal outliers, ensuring 92% signal accuracy.

U: Usage Trend (50%)

Weight is heavy as behavior is the truest churn predictor. Tracks session duration & feature depth.

S: Support Sentiment (30%)

Score derived from ticket volume, urgency, and automated NLP sentiment of ticket text.

E: Engagement (20%)

Responsive velocity to outreach and decision-maker (Admin) login frequency.

** Weights optimized via back-testing against churned account archetypes.*

THE NBA ENGINE

Prescriptive Intelligence

A score alone creates anxiety. The "Next-Best-Action" (NBA) engine transforms that score into a task. We mapped specific decay patterns to proven playbooks.

Automation Logic: NBA triggers are based on the specific vector of decay. We prioritize "High LTV" accounts in the Slack alert queue to optimize CSM response latency.

SCENARIO A: Usage drop in Reporting

NBA: AI-Drafted Training Invite

Goal: Remind Admin of ROI via automated visualization check.

SCENARIO B: Ticket backlog > 5 days

NBA: CS Director Check-in

Goal: Escalation to show care for recurring friction points.

SCENARIO C: Admin Inactive 10 days

NBA: Quarterly Review Prompt

Goal: Re-align with buyer on core business objectives.

Trigger Latency: < 2 Hours from Behavioral Shift Detection

05. UI/UX STRATEGY

Designing for Action

The design philosophy was **"Information to Action in < 3 Seconds."** We reduced cognitive load by hiding complexity and surfacing outcomes.

Design System: Leveraged a "Traffic Light" visual language. 92% of pilot users preferred 'Risk Velocity' sorting over legacy alphabetical lists.

> *"Arjun told us that he didn't care about the 20 variables. He just wanted a button that said 'Fix this account'."*

Risk Leaderboard

Accounts sorted by 'Risk Velocity' (how fast the score is dropping) rather than just static depth.

Evidence Card

Why is the score low? 3 clear proof points (e.g., 'API Integration Lag') to build trust in AI logic.

One-Click Recovery

Integrated email modal with context-aware drafts. Removes the 15-minute 'writer's block' for CSMs, allowing for 4x more outreach.

User Interaction Testing: 15 Iterative Cycles with CS Team

06. STRATEGIC TRADEOFFS

Behavior vs. NPS

Many stakeholders wanted NPS to be 50% of the score. I pushed back, arguing that **Behavior > Sentiment**. People lie on surveys; they don't lie with their mouse clicks. We kept NPS as a 'Context' field, not a 'Risk' field.

Decision Impact: Reduced false positives by 34% compared to NPS-only models.

Human-in-the-Loop

We vetoed "Full Automation" where AI emails the customer directly. High-ticket B2B relationships rely on brand trust. AI drafts the email, but the **CSM provides the soul** and the final 'Send' click.

Brand Safety: Zero complaints of "robotic" outreach recorded during the pilot.

Logic-First vs. ML-First

Instead of spending 4 months building a black-box ML model, we launched with a **Weighted Rule-Based Engine**. This gave CSMs transparent, explainable scores immediately and allowed us to collect clean training data for a true ML V2.

Strategic Aim: Transition to full neural-net prediction planned for Q4 after data enrichment.

07. IMPACT & ROI ANALYSIS

Financial Performance	Current (Pilot)	Post-Implementation
Monthly Revenue Leak	₹35.0 Lakh	₹27.5 Lakh
Monthly Churn %	7.0%	5.5%
Risk Audit Time / CSM	90 mins / day	**10 mins / day**
Annual Revenue Protected	**₹90,00,000**	

Assumptions: 1.5% Absolute Churn Reduction | 3-Year Projected Value: ₹2.7 Crore | ARR Growth Factor Excluded

08. REFLECTIONS & ROADMAP

Key Lesson Learned

Visibility ≠ Action. Just showing a "Red" account wasn't enough to change results. The breakthrough happened when we removed the cognitive load of "What do I do next?" by adding the **Suggested Action** button.

Team Impact: Reported morale boost in CS team as they felt like strategic advisors rather than clerical auditors.

V2 Roadmap

01 Reinforcement learning to refine playbooks based on success loops.

02 Zoom/Gong integration for NLP sentiment on customer calls.

03 Billing recovery automation for "payment-at-risk" triggers.

Next Steps: Secure Q4 Budget for Data Enrichment Phase

Thank You!

I'm happy to discuss the technical architecture or the user research methodology in further detail.

Portfolio: case-study-ai-retention.pdf

Contact: [yourportfolio.com]

IMAGE SOURCES

https://mokkupai.s3.ap-south-1.amazonaws.com/files/assets/blogs/cover-images/193.png

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